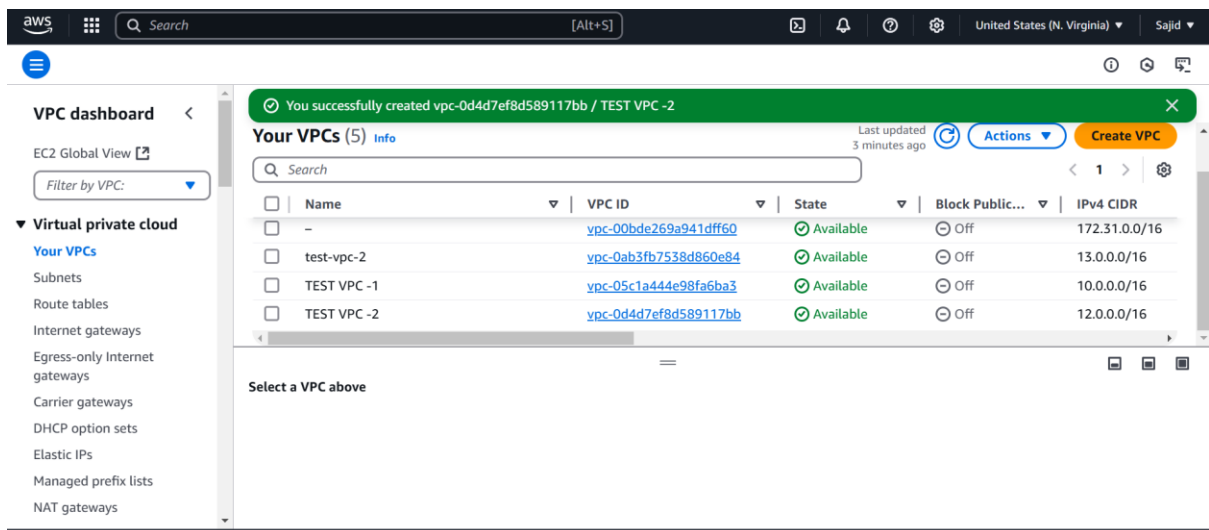


VPC PEERING CONNECTION

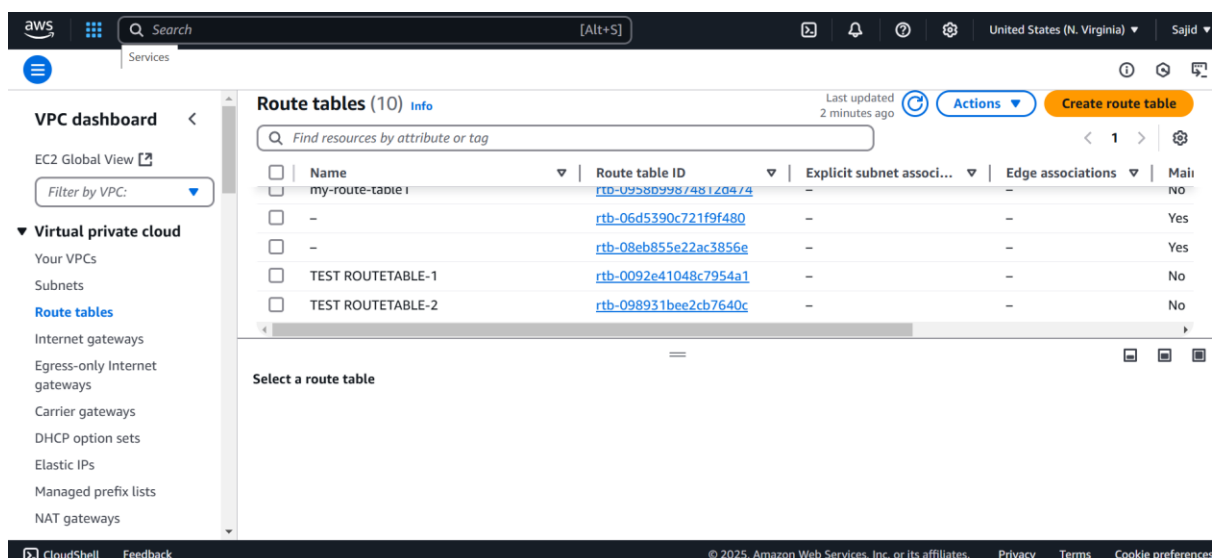
A VPC Peering connection is a networking connection between two Virtual Private Clouds (VPCs) that allows them to route traffic between each other using private IP addresses. VPC peering enables communication between VPCs within the same region or across different regions.

STEPS FOR VPC PEERING CONNECTION:

- Go to the AWS console
- Create two VPCs TEST VPC -1 , TEST VPC -2

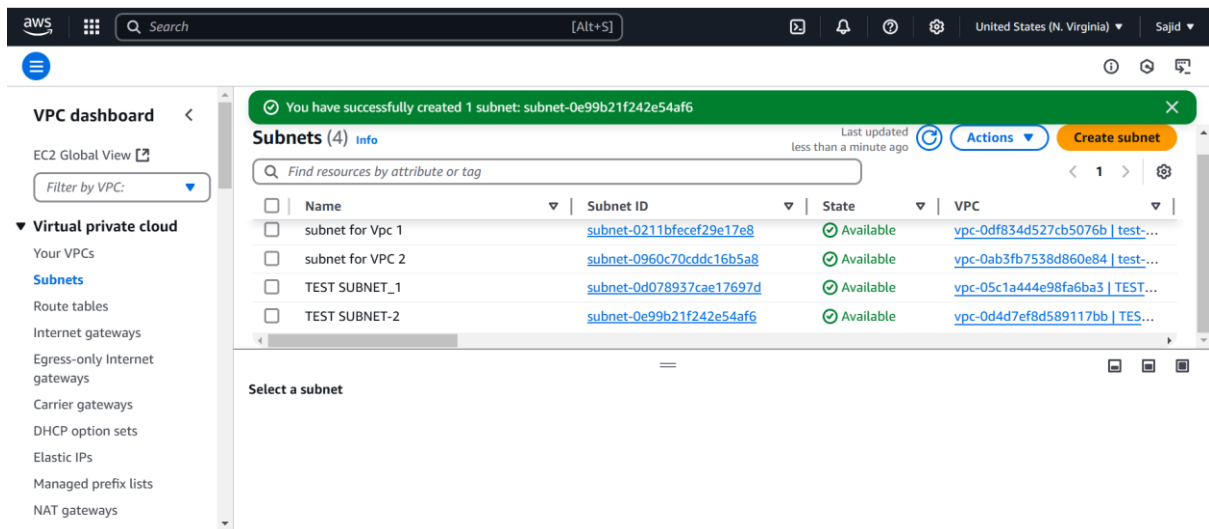


- Create two route tables TEST ROUTETABLE-1 & TEST ROUTETABLE-2.
- Attach the TEST ROUTETABLE-1 with TEST VPC-1
- Attach the TEST ROUTETABLE-2 With TEST VPC-2

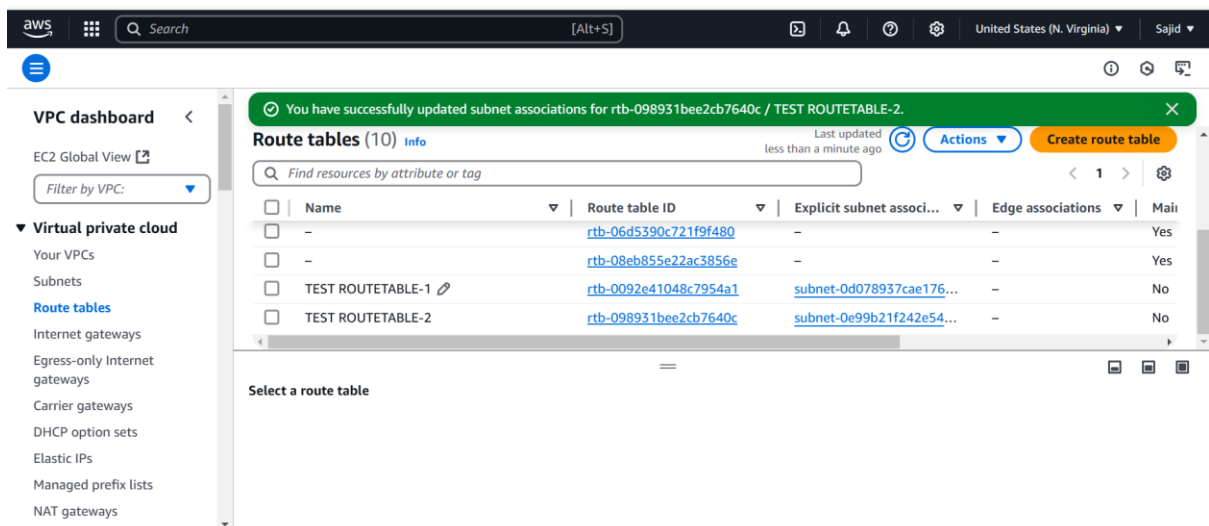


- CREATION OF SUBNETS

- Create two subnets TEST SUBNET-1 & TEST SUBNET-2
- Attach the TEST SUBNET-1 with TEST VPC-1
- Attach the TEST SUBNET-2 with TEST VPC-2



- Now associate the TEST ROUTETABLE-1 with TEST SUBNET-1
- And associate TEST ROUTETABLE-2 with TEST SUBNET-2



- Create Two Internet Gateways TEST IGW VPC-1 & TEST IGW VPC-2
- Attach TEST IGW VPC-1 with TEST VPC-1 & TEST IGW VPC-2 with TEST VPC-2.

Internet gateway igw-0e0254416264a1898 successfully attached to vpc-05c1a444e98fa6ba3

Internet gateways (1/3) Info

Name	Internet gateway ID	State	VPC ID
-	igw-0862927c21a0db2cc	Attached	vpc-00bde269a941dff60
igw for vpc 1	igw-0c3cbbce1dd4a28c0	Attached	vpc-0df834d527cb5076b tes
TEST IGW VPC-1	igw-0e0254416264a1898	Attached	vpc-05c1a444e98fa6ba3 TES

igw-0e0254416264a1898 / TEST IGW VPC-1

Details Tags

Details

Internet gateway ID	State	VPC ID	Owner
igw-0e0254416264a1898	Attached	vpc-05c1a444e98fa6ba3	878126318499

- Now route the TEST ROUTE TABLE-1 with TEST IGW VPC-1.

Route tables > rtb-0092e41048c7954a1 > Edit routes

Edit routes

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
0.0.0.0/0	Internet Gateway	-	No

Remove

Cancel Preview Save changes

- Route the TEST ROUTE TABLE-2 with TEST IGW VPC-2

Updated routes for rtb-098931bee2cb7640c / TEST ROUTETABLE-2 successfully

Details Info

Route table ID	Main	Explicit subnet associations	Edge associations
rtb-098931bee2cb7640c	No	subnet-0e99b21f242e54af6 / TEST SUBNET-2	-

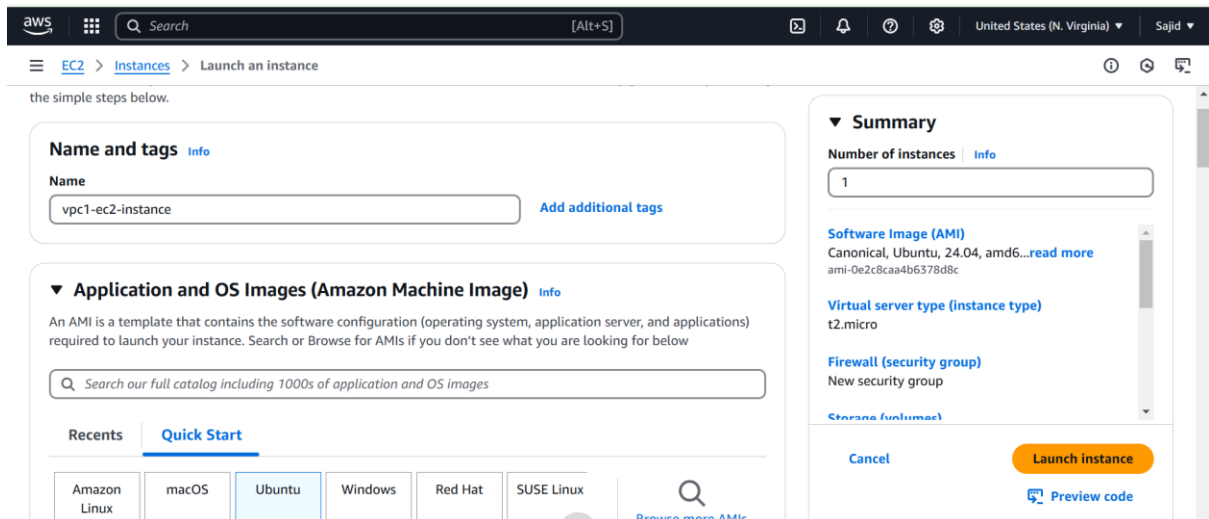
VPC vpc-0d4d7ef8d589117bb | TEST VPC-2

Owner ID 878126318499

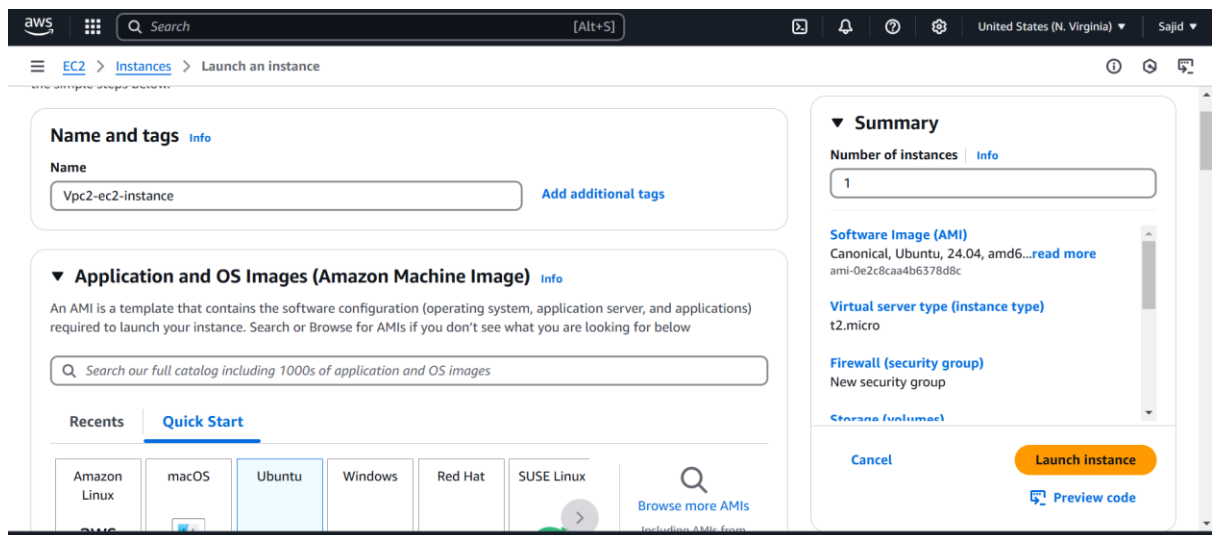
Routes (2)

Destination	Target	Status	Propagated
0.0.0.0/0	igw-07a8ae0870dd2ef91	Active	No
12.0.0.0/16	local	Active	No

- Create one Ec2 server in each VPC
- Create vpc1-ec2-instance with TEST VPC-1



- Create vpc2-ec2-instance with TEST VPC-2



- Connect to both the instances.

STEPS TO CONNECTON BETWEEN TWO EC2 SERVERS IN DIFFERENT VPCs:

- Go to VPC
- Click on peering connection.
- Click on create peering connection
- Give the name as connection between TEST VPC-1 & TEST VPC-2
- Select the requester VPC (TEST VPC-1) & Acceptor VPC (TEST VPC-2)

VPC > Peering connections > Create peering connection

Select a local VPC to peer with

VPC ID (Requester)
vpc-05c1a444e98fa6ba3 (TEST VPC -1)

VPC CIDRs for vpc-05c1a444e98fa6ba3 (TEST VPC -1)

CIDR	Status	Status reason
10.0.0.0/16	Associated	-

Select another VPC to peer with

Account
☒ My account
☐ Another account

Region
☒ This Region (us-east-1)
☐ Another Region

VPC ID (Accepter)
vpc-0d4d7ef8d589117bb (TEST VPC -2)

- Click on create peering connection.

aws Search [Alt+S] United States (N. Virginia) Sajid

VPC > Peering connections > pcx-0fe7407645391a17f

VPC dashboard

EC2 Global View

Filter by VPC:

Virtual private cloud

- Your VPCs
- Subnets
- Route tables
- Internet gateways
- Egress-only Internet gateways
- Carrier gateways
- DHCP option sets
- Elastic IPs
- Managed prefix lists
- NAT gateways

A VPC peering connection pcx-0fe7407645391a17f / connection between TEST VPC-1 & TEST VPC-2 has been requested.

pcx-0fe7407645391a17f / connection between TEST VPC-1 & TEST VPC-2 Actions

Pending acceptance
You can accept or reject this peering connection request using the 'Actions' menu. You have until Friday 17 January 2025 at 11:54:17 GMT+5:30 to accept or reject the request, otherwise it expires.

Details

Requester owner ID 878126318499	Accepter owner ID 878126318499	VPC Peering connection ARN arn:aws:ec2:us-east-1:878126318499:vpc-peering-connection/pcx-0fe7407645391a17f
Peering connection ID pcx-0fe7407645391a17f	Requester VPC vpc-05c1a444e98fa6ba3 / TEST VPC -1	Accepter VPC vpc-0d4d7ef8d589117bb / TEST VPC -2
Status Pending Acceptance by 878126318499	Requester CIDRs 10.0.0.0/16	Accepter CIDRs -
Expiration time Friday 17 January 2025 at 11:54:17 GMT+5:30	Requester Region N. Virginia (us-east-1)	Accepter Region N. Virginia (us-east-1)

https://us-east-1.console.aws.amazon.com/vpcconsole/home?region=us-east-1#Home: © 2025, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences

- Now we need to accept or reject the peering connection request
- Click on Actions and Accept the request.

The first screenshot shows the 'Pending acceptance' state of a VPC peering connection. A green notification banner at the top states: 'A VPC peering connection pcx-0fe7407645391a17f / connection between TEST VPC-1 & TEST VPC-2 has been requested.' Below this, the connection details are displayed:

- Requester owner ID:** 878126318499
- Peering connection ID:** pcx-0fe7407645391a17f
- Status:** Pending Acceptance by 878126318499
- Expiration time:** Friday 17 January 2025 at 11:54:17 GMT+5:30
- Accepter owner ID:** 878126318499
- Requester VPC:** vpc-05c1a444e98fa6ba3 / TEST VPC -1
- Requester CIDRs:** 10.0.0.0/16
- Requester Region:** N. Virginia (us-east-1)
- VPC Peering connection ARN:** arn:aws:ec2:us-east-1:878126318499:vpc-peering-connection/pcx-0fe7407645391a17f
- Accepter VPC:** vpc-0d4d7ef8d589117bb / TEST VPC -2
- Accepter CIDRs:** -
- Accepter Region:** N. Virginia (us-east-1)

The 'Actions' menu is open, showing options: 'Accept request', 'Reject request', 'Edit DNS settings', 'Manage tags', and 'Delete peering connection'.

The second screenshot shows the 'Active' state of the same VPC peering connection. A green notification banner at the top states: 'Your VPC peering connection (pcx-0fe7407645391a17f | connection between TEST VPC-1 & TEST VPC-2) has been established. To send and receive traffic across this VPC peering connection, you must add a route to the peered VPC in one or more of your VPC route tables.' Below this, the connection details are updated:

- Status:** Active
- Accepter CIDRs:** 12.0.0.0/16

- Now we need to modify the routetables of the TEST VPC-1 & TEST VPC-2
- Go to the route tables
- Click on TEST ROUTETABLE-1
- Click on routes
- Click on edit routes.
- Add route to TEST VPC-2 by using IPv4 of TEST VPC-2.

aws [Search] [Alt+S] United States (N. Virginia) Sajid

VPC > Route tables > rtb-0092e41048c7954a1 > Edit routes

Edit routes

Destination	Target	Status	Propagated	
10.0.0.0/16	local	Active	No	
0.0.0.0/0	Internet Gateway	Active	No	Remove
12.0.0.0/16	Peering Connection	-	No	Remove

Add route

Cancel Preview Save changes

- Click on save changes.

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VPC > Route tables > rtb-0092e41048c7954a1

Updated routes for rtb-0092e41048c7954a1 / TEST ROUTETABLE-1 successfully

rtb-0092e41048c7954a1 / TEST ROUTETABLE-1

Details Info

Route table ID rtb-0092e41048c7954a1	Main No	Explicit subnet associations subnet-0d078937cae17697d / TEST SUBNET_1	Edge associations -
VPC vpc-05c1a444e98fa6ba3 TEST VPC -1	Owner ID 878126318499		

Routes (3)

Filter routes

Destination	Target	Status	Propagated
10.0.0.0/16	local	Active	No
0.0.0.0/0	Internet Gateway	Active	No
12.0.0.0/16	Peering Connection	-	No

Both Edit routes

- Now click on the TEST ROUTE TABLE-2
- Click on routes
- Click on edit routes
- Add route to TEST VPC-1 by using IPv4 of the TEST VPC-1

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VPC > Route tables > rtb-098931bee2cb7640c > Edit routes

Edit routes

Destination	Target	Status	Propagated	
12.0.0.0/16	local	Active	No	
0.0.0.0/0	Internet Gateway	Active	No	Remove
10.0.0.0/16	Peering Connection	-	No	Remove

Add route

Cancel Preview Save changes

- Click on save changes

Updated routes for rtb-098931bee2cb7640c / TEST ROUTETABLE-2 successfully

rtb-098931bee2cb7640c / TEST ROUTETABLE-2

Details Info

Route table ID rtb-098931bee2cb7640c	Main No	Explicit subnet associations subnet-0e99b21f242e54af6 / TEST SUBNET-2	Edge associations -
VPC vpc-0d4d7ef8d589117bb TEST VPC -2	Owner ID 878126318499		

Routes (3)

Destination	Target	Status	Propagated
0.0.0.0/0	igw-0a1b2c3d4e5f6g7h	Complete	Yes
10.0.0.0/24	subnet-0e99b21f242e54af6	Complete	Yes
10.0.1.0/24	subnet-0e99b21f242e54af6	Complete	Yes

- Connect to the both instances
- Run the command “curl private ip” of the vpc2-ec2-instance in the vpc1-ec2-instance.
- You are able to access the” vpc2-ec2-instance” server through “ vpc1-ec2-instance”.